CAUSE & EFFECT MATRIX — LCP-01 Scrubber & Burner Control

Project: Burner PetroBaker | Panel: LCP-01 | Ref. drawing: PH-LCP-01 V1 | Document: PH-LCP-01-CE | Rev. A (for client review) | 26-10-2026

Purpose	Defines the PLC (LCP-01) response to every process and safety cause, as requested in the client e-mail of 04-Oct-2026 (Eng. Amr Farag).
Scope	Gas scrubber alarms/trips, ESD, burner BR01 (Riello RS100) and BR02 (DLX 53 SHFP) trips and permissives, tank temperature control (client sheet 'Temperature control of burner'). This matrix lists causes and effects only. Operating modes (AUTO / MAN), START / STOP, RESET, start-up after power loss and lamps are described in the Control Narrative (docs/التشغيل.md).
Not in scope (Rev. A)	Communication with the Master PLC in the control room — deferred to a later revision. Scrubber low level switch — not installed (DI I12.5 reserved).
Burner flame safety	Purge, ignition, flame supervision and the pilot / main gas valves are handled by each burner's own certified control box (RS100: RMG88; DLX 53: SQ DUAL). LCP-01 only gives the burner its heat request (TL / L) and stage request (TR / T2), and reads its RUN and LOCKOUT signals. LCP-01 never drives a gas valve directly.
Hardwired safety	E-STOP S0 opens the 230 V control supply of BR01 (NC2) and BR02 (NC3) directly, independent of the PLC. Safety thermostat TS (manual reset) is in series with the BR01 control supply. These act even if the PLC fails.
Fail-safe outputs	All PLC outputs de-energise on PLC STOP or 24 VDC loss. Valve types were chosen so that de-energised = safe: XV-401 flare (NO) OPENS, XV-402 drain (NO) OPENS, XV-403 suction (NC) CLOSES, both burners lose their heat request.
How to read the matrix	One row per cause. Cells under each effect say what happens to that output while the cause is active. Empty = no action. STOP = output de-energised / burner request removed. RUN = energised. OPEN / CLOSE = resulting valve position. ALARM = HMI alarm (the burner fault lamps H3 / H4 are wired to the burner lockout, not to the PLC). Type: Trip = stops and stays stopped until RESET (S2). Permissive = stops, restarts by itself when the condition clears (AUTO mode, burner enabled). Control = normal temperature regulation. Alarm = HMI message only. Inherent = happens by wiring, no program. Column 'Source': Client = client document; PH decision = decided by PH (Eng. Ahmed Zain); PH proposal = proposed by PH, still open.
Requested from client	On sheet 'C&E Matrix', fill the two yellow columns for every row: 'Client response' (Approved / Approved with comment / Rejected) and 'Client comment'. Rows marked 'PH proposal' need particular attention. All other cells are protected against accidental edits (no password). Use the filter on the header row, e.g. Source = PH proposal. Parameter values are on sheet 'Parameters' and are adjustable from the HMI.

Revision	Date	Description	By / Approved
A	26-10-2026	First issue for client review	PH / Eng. Ahmed Zain

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No.	Tag	Cause	PLC address	Condition / setpoint	Type	EFFECTS								Reset	Source	Remarks	Client response	Client comment
						E1 BR01 heat request (TL)	E2 BR01 2nd stage (TR)	E3 BR02 start (L)	E4 BR02 high fire (T2)	E5 XV-401 Flare SV (NO)	E6 XV-402 Drain SV (NO)	E7 XV-403 Suction SV (NC)	E8 HMI alarm					
						Q0.0 / R1	Q0.1 / R2	Q0.5 / R6	Q0.6 / R7	Q0.2 / R3	Q0.3 / R4	Q0.4 / R5	HMI					
EMERGENCY SHUTDOWN																		
C01	S0	E-STOP pressed	I1.0	NC1 contact open	Trip	STOP	STOP	STOP	STOP	OPEN	OPEN	CLOSE	ALARM	Release S0, then S2; burners need START	PH proposal	Burners are also cut by hardwired S0 NC2/NC3 (client e-mail: ESD impact on burner ON/OFF). Scrubber valves to safe position on ESD = PH proposal.		
C02	—	PLC STOP, 24 VDC loss or power-up	—	All outputs de-energised	Inherent	STOP	STOP	STOP	STOP	OPEN	OPEN	CLOSE		S2 for scrubber; burners need START	PH proposal	Result of valve types (NO / NO / NC) and relay outputs. After power returns nothing restarts by itself.		
GAS SCRUBBER																		
C03	PSH-101	Scrubber high pressure	I0.0	Contact open (fail-safe)	Trip					OPEN	OPEN	CLOSE	ALARM	S2 after cleared	PH decision	Client e-mail asks for scrubber alarms/trips; the valve action and burner policy were decided by PH. Burners are NOT tripped (decision PH / Eng. Ahmed Zain); if gas pressure to the burner drops, the burner's own gas pressure switch stops it (C07g / C08g).		
C04	LSH-102	Scrubber high level	I0.1	Contact open (fail-safe)	Alarm								ALARM	Automatic when cleared	Client	Pre-alarm before LSHH-103.		
C05	LSHH-103	Scrubber high-high level	I0.2	Contact open (fail-safe)	Trip					OPEN	OPEN	CLOSE	ALARM	S2 after cleared	PH decision	Burners are NOT tripped (as C03).		
C06	LSL (scrubber)	Scrubber low level	I12.5 (reserved)	—	Not installed									—	PH decision	Provisional: understood as not installed (open question Q1). Input reserved.		
BURNER BR01 — tank TK01 (operating modes, START / STOP: see Control Narrative)																		
C07a	LSL-TK01	TK01 level below low level switch	I1.2	LSL not reached (contact open)	Permissive	STOP	STOP							Automatic when level returns	Client	Client sheet: burner may run only when level has reached LSL. Contact closes with liquid present, so a broken wire also blocks the burner.		
C07b	TE-501 / TE-502	TK01 oil temperature at setpoint	IW96 / IW98	max(T) >= SP (70 °C)	Control	STOP	STOP							Automatic at max(T) <= SP − HYST (65 °C)	Client	Client cases 2/3/6. The higher of inlet/outlet is used and the 5 °C restart band are PH proposals: each restart costs ~50 s of purge.		
C07c	TE-501 / TE-502	TK01 oil temperature low — fast heat-up	IW96 / IW98	max(T) < SP − STG (60 °C)	Control		RUN							Automatic at max(T) >= 60 °C	PH proposal	2nd stage / high fire. Between 60 and 70 °C the burner runs on 1st stage.		
C07d	TE-501 / TE-502	Thermocouple fault — either of the two	IW96 / IW98	Either channel: 32767 / −32768, outside −20…150 °C, or channel diagnostic	Trip	STOP	STOP						ALARM	S2 after fault cleared	PH proposal	Each channel is checked BEFORE taking the higher value, so one failed-low sensor cannot hide behind the other. Heat is inhibited until both channels are valid after power-up.		
C07e	Rflt	BR01 lockout (flame / ignition / air pressure failure)	I0.5	Lockout signal present	Trip	STOP	STOP						ALARM	At burner control box, then S2	PH proposal	The burner's own control box has already closed the gas valves. PLC removes the heat request so the burner does not restart by itself after a local reset.		
C07f	RR	BR01 failed to start	I0.3	Heat request ON, no RUN within T_START (120 s)	Trip	STOP	STOP						ALARM	S2	PH proposal	RUN = gas valves energised (firing). Normal start ~55 s (RS100 §6.8.1) / ~65 s (DLX 53).		
C07g	RR	BR01 stopped by itself while running	I0.3	RUN lost, no lockout, heat request still ON	Trip	STOP	STOP						ALARM	S2	PH proposal	E.g. safety thermostat TS, gas pressure switch, burner supply lost. Without this row the PLC would keep requesting heat from a burner that is no longer firing.		
C07h	—	BR01 running continuously for 24 h	—	Run timer >= 24 h	Permissive	STOP	STOP							Automatic (restarts on next demand)	PH proposal	RS100 is approved for intermittent operation and must stop at least once every 24 h (RS100 manual §5.10).		
BURNER BR02 — tank TK02 (operating modes, START / STOP: see Control Narrative)																		
C08a	LSL-TK02	TK02 level below low level switch	I1.3	LSL not reached (contact open)	Permissive			STOP	STOP					Automatic when level returns	Client	Client sheet: burner may run only when level has reached LSL. Contact closes with liquid present, so a broken wire also blocks the burner.		
C08b	TE-503 / TE-504	TK02 oil temperature at setpoint	IW100 / IW102	max(T) >= SP (70 °C)	Control			STOP	STOP					Automatic at max(T) <= SP − HYST (65 °C)	Client	Client cases 2/3/6. The higher of inlet/outlet is used and the 5 °C restart band are PH proposals: each restart costs ~50 s of purge.		
C08c	TE-503 / TE-504	TK02 oil temperature low — fast heat-up	IW100 / IW102	max(T) < SP − STG (60 °C)	Control				RUN					Automatic at max(T) >= 60 °C	PH proposal	2nd stage / high fire. Between 60 and 70 °C the burner runs on 1st stage.		
C08d	TE-503 / TE-504	Thermocouple fault — either of the two	IW100 / IW102	Either channel: 32767 / −32768, outside −20…150 °C, or channel diagnostic	Trip			STOP	STOP				ALARM	S2 after fault cleared	PH proposal	Each channel is checked BEFORE taking the higher value, so one failed-low sensor cannot hide behind the other. Heat is inhibited until both channels are valid after power-up.		
C08e	Rflt2	BR02 lockout (flame / ignition / air pressure failure)	I1.5	Lockout signal present	Trip			STOP	STOP				ALARM	At burner control box, then S2	PH proposal	The burner's own control box has already closed the gas valves. PLC removes the heat request so the burner does not restart by itself after a local reset.		
C08f	RR2	BR02 failed to start	I1.4	Heat request ON, no RUN within T_START (120 s)	Trip			STOP	STOP				ALARM	S2	PH proposal	RUN = gas valves energised (firing). Normal start ~55 s (RS100 §6.8.1) / ~65 s (DLX 53).		
C08g	RR2	BR02 stopped by itself while running	I1.4	RUN lost, no lockout, heat request still ON	Trip			STOP	STOP				ALARM	S2	PH proposal	E.g. safety thermostat TS, gas pressure switch, burner supply lost. Without this row the PLC would keep requesting heat from a burner that is no longer firing.		

ADJUSTABLE PARAMETERS (HMI)

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Parameter	Description	Default	Unit	Range	Source
SP	Tank temperature setpoint — burner stops at T >= SP	**70**	°C	40 – 90	Client sheet (adjustable from SCADA/HMI)
HYST	Restart hysteresis — burner restarts at T <= SP − HYST	**5**	°C	3 – 10	PH proposal: > type-K accuracy (±2.2 °C), limits short-cycling
STG	2nd stage / high fire while T < SP − STG	**10**	°C	5 – 30	PH proposal
T_START	Max. time from heat request to RUN feedback	**120**	s	90 – 300	PH proposal: ~2 × burner start cycle
T_MAXRUN	Forced stop of BR01 after continuous run	24	h	fixed	RS100 manual §5.10

Blue values are defaults; they can be changed from the HMI by an authorised user.

I/O LIST (from drawing PH-LCP-01 V1, sheets 10–13)

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Address	Tag	Description	Terminal	Signal	Sheet
I0.0	PSH-101	Scrubber high pressure	XI:1	DI 24V	10
I0.1	LSH-102	Scrubber high level	XI:3	DI 24V	10
I0.2	LSHH-103	Scrubber high-high level	XI:5	DI 24V	10
I0.3	RR	BR01 run feedback	XI:7	DI 24V	10
I0.4	S1	BR01 START push button	XI:9	DI 24V	10
I0.5	Rflt	BR01 fault (lockout)	XI:11	DI 24V	10
I0.6	S3	BR01 STOP push button	XI:13	DI 24V	10
I0.7	—	Spare	XI:15	DI 24V	10
I1.0	S0	E-STOP status (NC1)	XI:17	DI 24V	10
I1.1	S2	FAULT RESET push button	XI:19	DI 24V	10
I1.2	LSL-TK01	TK01 low level	XI:21	DI 24V	10
I1.3	LSL-TK02	TK02 low level	XI:23	DI 24V	10
I1.4	RR2	BR02 run feedback	XI:25	DI 24V	10
I1.5	Rflt2	BR02 fault (lockout)	XI:27	DI 24V	10
I12.0	Rfuel2	BR02 diesel mode	XI2:1	DI 24V	11
I12.1	S4	BR01 AUTO/MAN selector (closed = AUTO)	XI2:3	DI 24V	11
I12.2	S5	BR02 AUTO/MAN selector (closed = AUTO)	XI2:5	DI 24V	11
I12.3	S6	BR02 START push button	XI2:7	DI 24V	11
I12.4	S7	BR02 STOP push button	XI2:9	DI 24V	11
I12.5	—	Reserved: scrubber LSL (not installed)	XI2:11	DI 24V	11
I12.6–7	—	Spare	XI2:13–16	DI 24V	11
Q0.0	R1	BR01 heat request (TL) + H1 1st stage lamp	XQ:1/2	DQ 24V	12
Q0.1	R2	BR01 2nd stage (TR) + H2 2nd stage lamp	XQ:3/4	DQ 24V	12
Q0.2	R3	XV-401 Flare SV (NO)	XQ:5/6	DQ 24V	12
Q0.3	R4	XV-402 Drain SV (NO)	XQ:7/8	DQ 24V	12
Q0.4	R5	XV-403 Suction SV (NC)	XQ:9/10	DQ 24V	12
Q0.5	R6	BR02 start (L)	X6:2	DQ 24V	12 / 8
Q0.6	R7	BR02 high fire (T2)	X6:4	DQ 24V	12 / 8
Q0.7	R8	BR02 1st stage lamp H5	—	DQ 24V	12
Q1.0	R9	BR02 2nd stage lamp H6	—	DQ 24V	12
Q1.1	R10	Spare	XQ:19/20	DQ 24V	12
IW96	TE-501	TK01 inlet oil temperature	X5:1/2	TC type K	13
IW98	TE-502	TK01 outlet oil temperature	X5:3/4	TC type K	13
IW100	TE-503	TK02 inlet oil temperature	X5:5/6	TC type K	13
IW102	TE-504	TK02 outlet oil temperature	X5:7/8	TC type K	13
IW104–110	—	Spare thermocouple channels	X5:9–16	TC	13